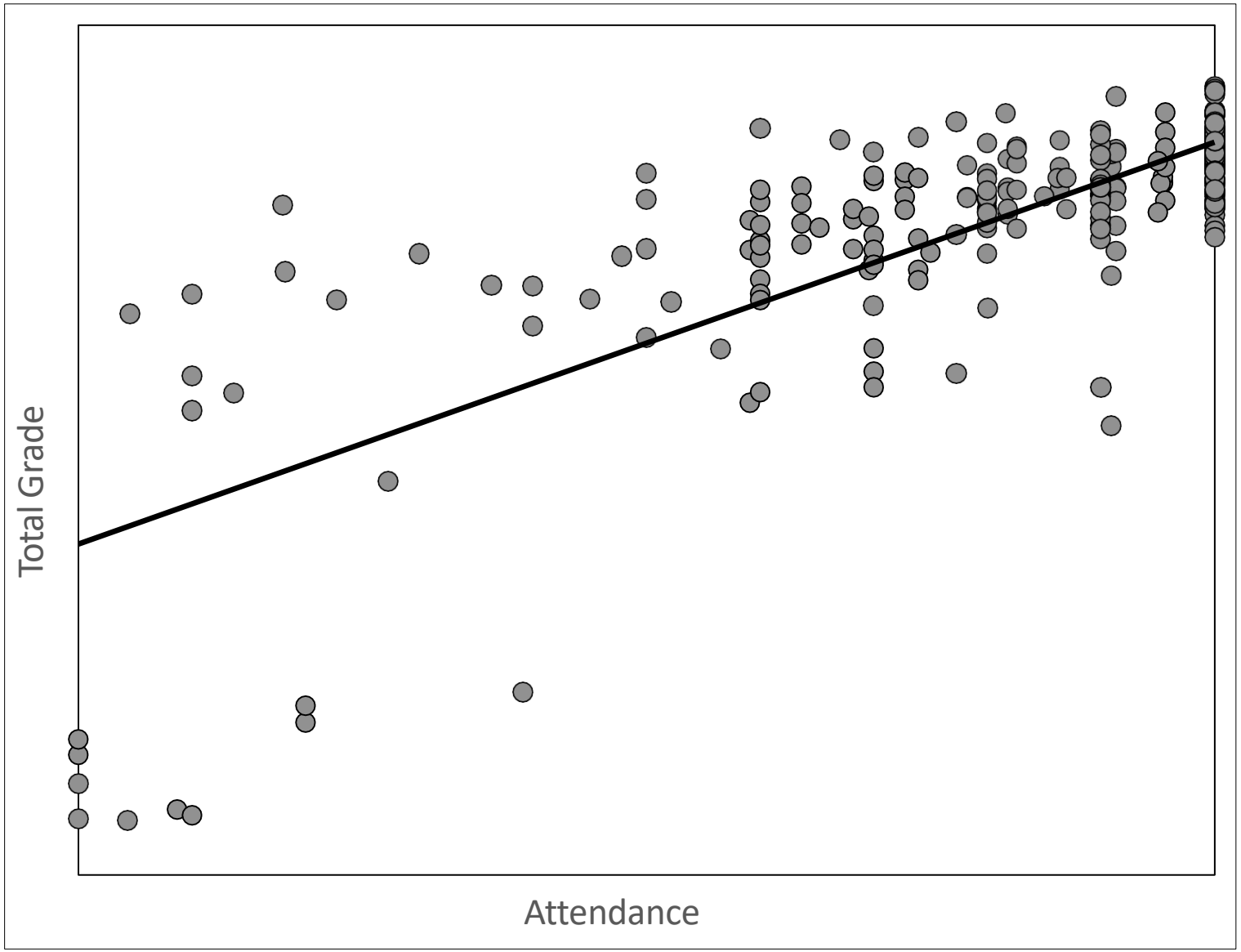
The Mechanical Engineering Profession

ME 110 Introduction to Mechanical Engineering

Lecture Slides #1


Total Grade
Attendance

Weekly Schedule

What's a University?		22 Eylül 2026
Engineering Profession	*Lecture 1*	*29 Eylül 2026*
Engineering Design	Lecture 2	06 Ekim 2026
Manufacturing	Lecture 3	13 Ekim 2026
Effective Writing Skills and Presentation	Lecture 4	20 Ekim 2026
Mechanics	Lecture 5	27 Ekim 2026
Project Planning Team Work	Lecture 6	03 Kasım 2026
Vize		07 - 15 Kasım 2026
Strength,Stress and Materials	Lecture 7	17 Kasım 2026
Practice		24 Kasım 2026
Ethics	Lecture 8	01 Aralık 2026
Project Presentation		08 Aralık 2026
Problem Solving Methods	Lecture 9	15 Aralık 2026
Dynamics, Mechanisms and Control	Lecture 10	22 Aralık 2026
Thermofluids and Energy	Lecture 11	29 Aralık 2026
Final		04 - 17 Ocak 2027
Bütünleme		19 - 24 Ocak 2027

What you will learn

By the end of this lecture, you should be able to:

- Describe the work of an engineer and a mechanical engineer
- Connect core courses to real machines and systems

What Is Engineering?

Engineering is the application of scientific and mathematical knowledge to the design of useful solutions.

- Engineers begin with a need or a problem.
- They work within constraints such as cost, available materials, time, safety, and environmental impact.
- They use calculations, models, experiments, and judgment to compare possible solutions.
- A successful design must perform its intended function under real operating conditions.

The Meaning of “Mühendis”

The Turkish word **mühendis** is associated with *hendese*, a historical term for geometry.

> This connection reflects the importance of measurement and calculation in engineering. Today, however, engineering involves much more than calculation. An engineer must also define problems, select materials and processes, evaluate risks, test designs, and communicate decisions.

Can a mathematically and physically correct design still be a poor engineering solution?

What Is Mechanical Engineering?

Mechanical engineering concerns the **design, analysis, manufacture, operation, and improvement of physical systems** involving forces, motion, materials, fluids, and energy.

Mechanical engineers work on products as small as medical devices and as large as power plants. Their work may involve a single component, such as a bearing, or a complete system, such as a vehicle or a steel production line.



Why Is Mechanical Engineering So Broad?

A machine rarely depends on only one physical principle.
Consider an electric vehicle:

- Its structure must carry loads.
- Its motor and transmission produce motion.
- Its battery and electronics require thermal management.
- Its components must be manufactured and assembled.
- Its behavior must be measured and controlled.

Mechanical engineering connects these questions into a functioning system.

Before Modern Industry

People developed mechanical devices long before mechanical engineering became a university discipline.

- Levers and pulleys changed the magnitude or direction of a force.
- Wheels and gears transmitted motion.
- Waterwheels and windmills converted natural energy into mechanical work.
- Pumps moved water for agriculture, mining, and settlements.

The enduring questions were practical: **How can we create motion, transmit power, and reduce human effort?**

The Industrial Revolution

Steam-powered machines expanded the availability of mechanical power. Factories increasingly depended on engines, shafts, gears, pumps, and machine tools.

This development created new engineering demands:

- Parts had to fit together reliably.
- Machines needed consistent materials and dimensions.
- Operators needed methods to inspect and maintain equipment.
- Designers needed ways to predict performance before construction.

Engines and Transportation

Heat engines helped transform transportation and industry by converting thermal energy into mechanical work.

The development of locomotives, ships, automobiles, and aircraft required more than an engine. Engineers also had to address structural loads, friction, cooling, fuel use, braking, vibration, and manufacturing.

Key idea: A major invention becomes useful at scale only when the complete system works reliably.

What Does a Mechanical Engineer Actually Do?

Mechanical engineers may:

- Define what a product or system must achieve.
- Calculate loads, temperatures, flow rates, and power requirements.
- Create drawings and select materials.
- Choose manufacturing and assembly methods.
- Test prototypes and investigate failures.
- Improve the performance, cost, safety, or service life of existing equipment.

The balance of these tasks depends on the job.

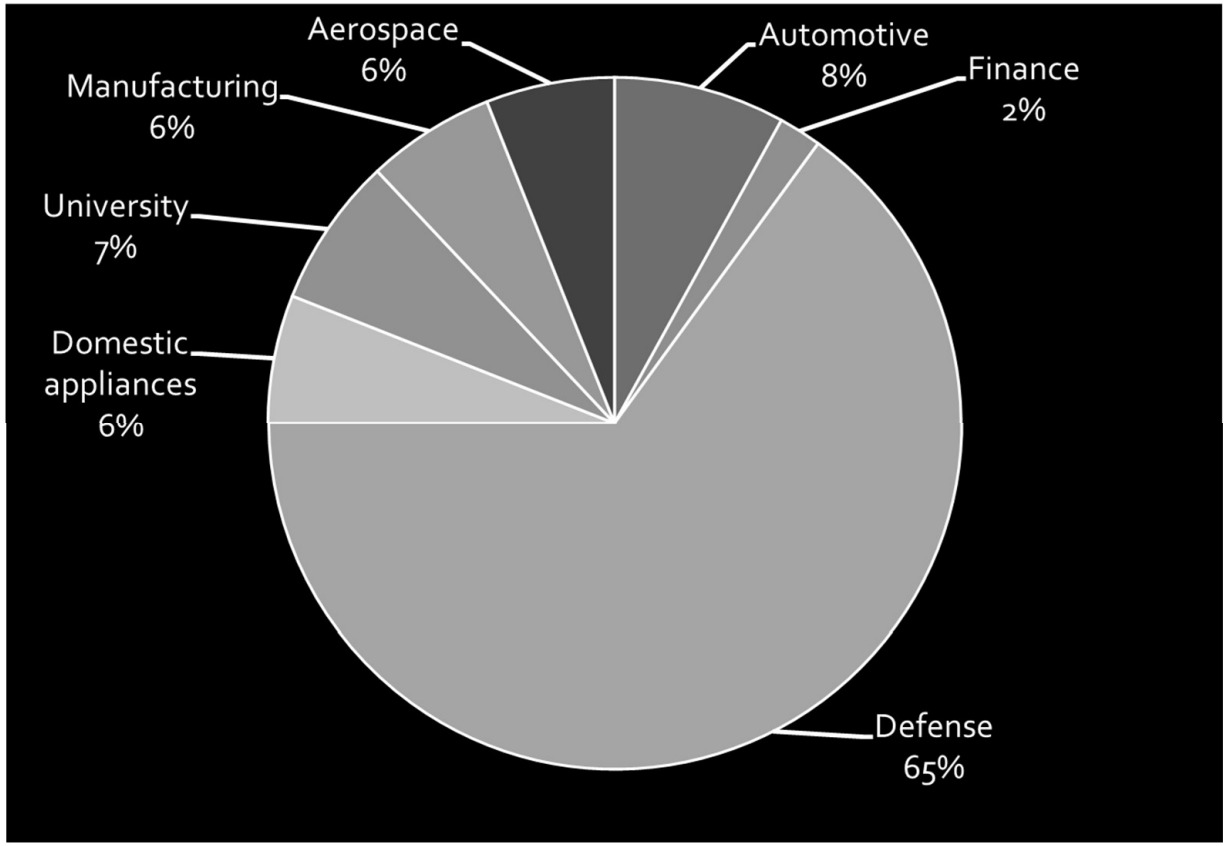
Scientists and Engineers

Scientists and engineers often use the same mathematics, instruments, and experiments, but their immediate questions may differ.

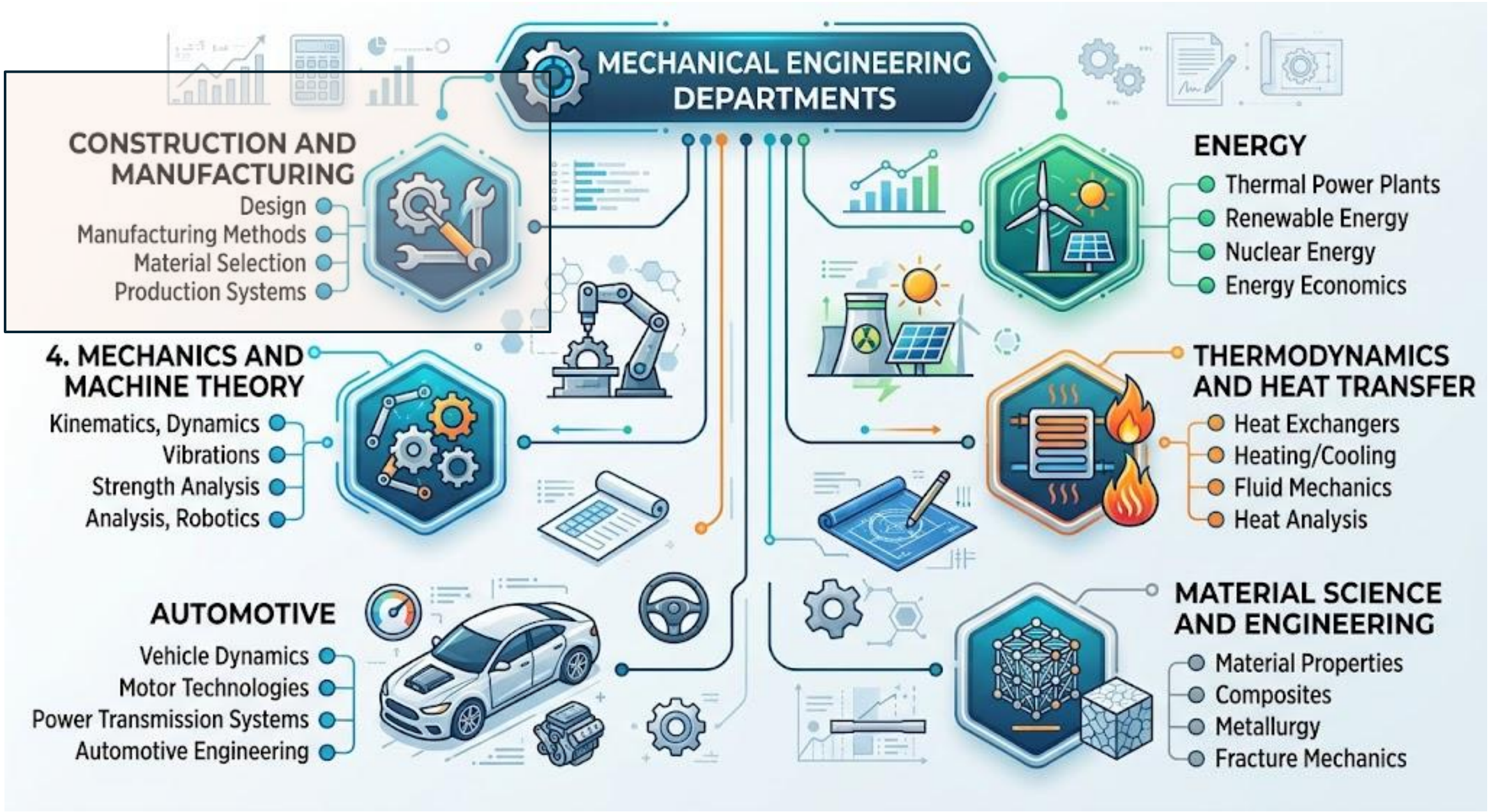
A scientist may ask: **Why does this material behave this way?**
An engineer may ask: **Can we use this material safely in this component?**

The distinction is not absolute. Engineers produce new knowledge through research, while scientific discoveries often depend on instruments designed by engineers.

Engineering Disciplines



Percentages of all METU Mechanical Engineering Department graduates working in the various engineering fields and their specializations.


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Mechanical Engineering in the Digital Age

Digital tools have changed how engineers develop machines.

- **CAD** describes the geometry of parts and assemblies.
- **CAE** helps predict structural, thermal, and fluid behavior.
- **CNC manufacturing** produces parts using computer-controlled machines.
- **Sensors** provide data from prototypes and operating equipment.

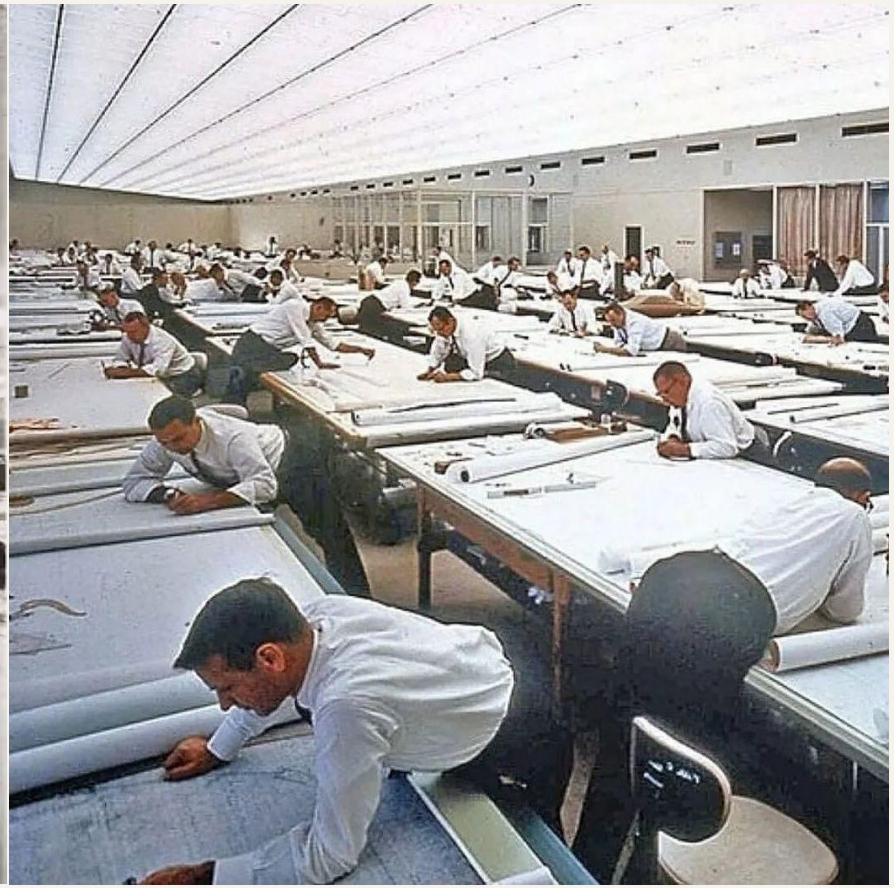
These tools accelerate development, but engineers must still understand their assumptions and compare predictions with measurements.

Mechanical Engineering in the Digital Age-CAD

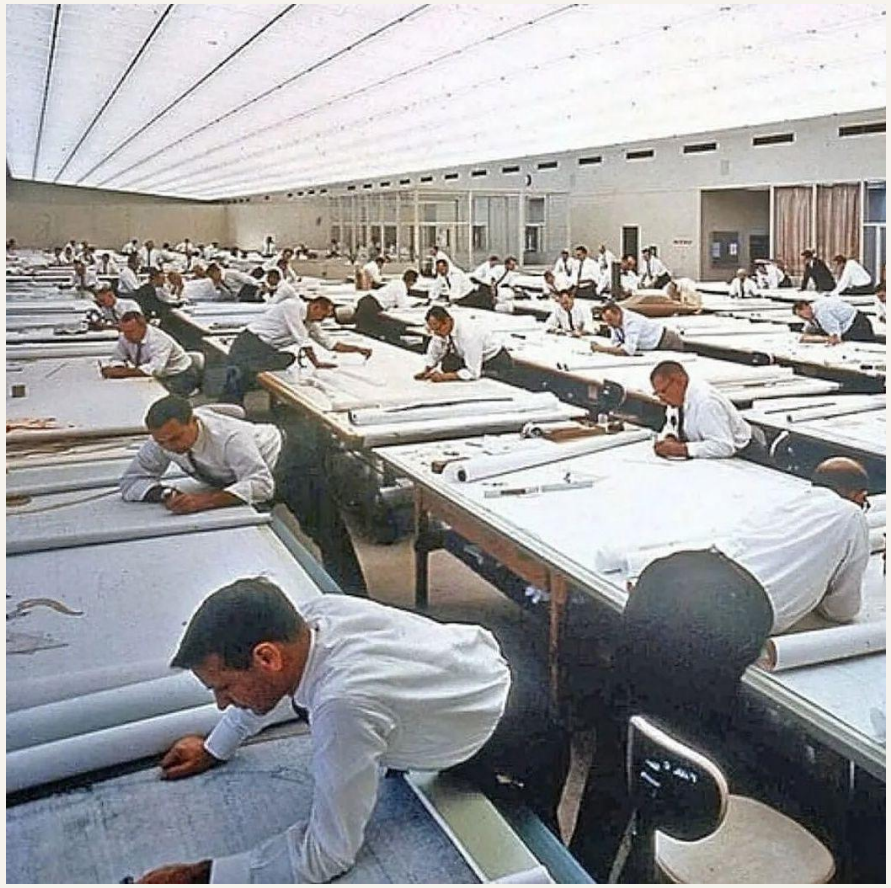
CAD (Computer Aided Drawing) – is the use of computer software to create, modify, analyze, or optimize a design. It is widely used by engineers, architects, and designers to produce 2D drawings and 3D models of products, buildings, or systems before they are built.

- Siemens NX
- SolidWorks
- CATIA
- AutoCAD
- Fusion 360
- Rhino
- ThinkerCAD
- Creo



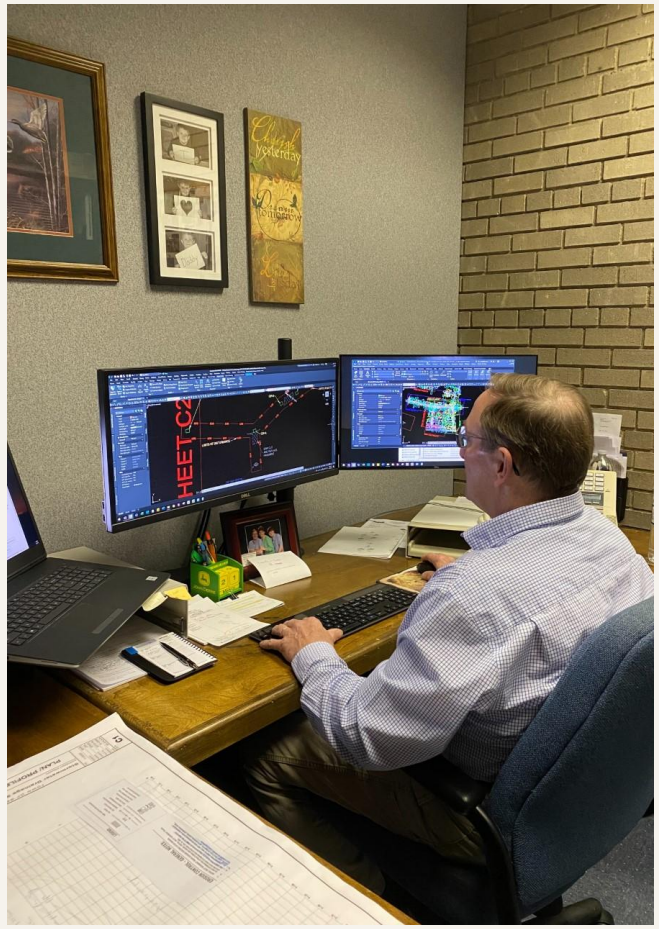


Mechanical Engineering in the Digital Age-CAD



Before

After

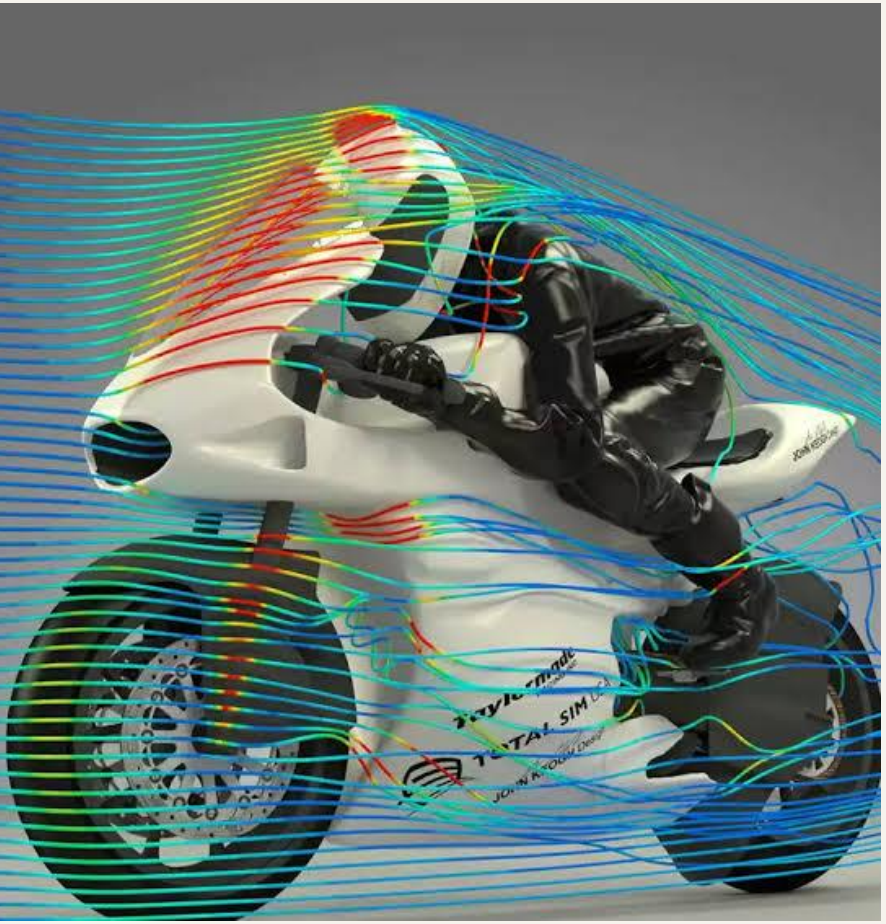


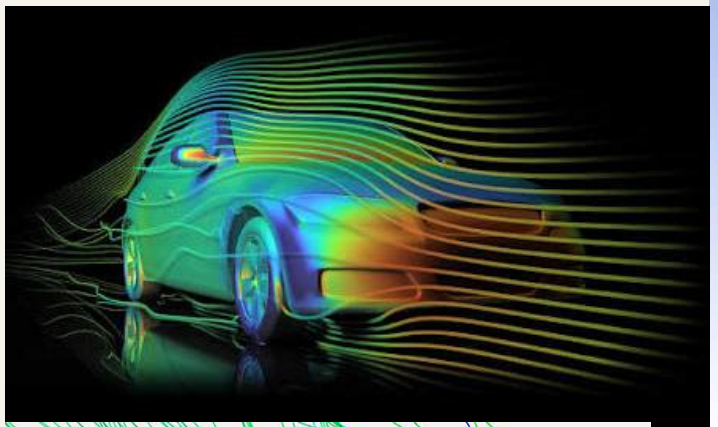
Mechanical Engineering in the Digital Age-CAE

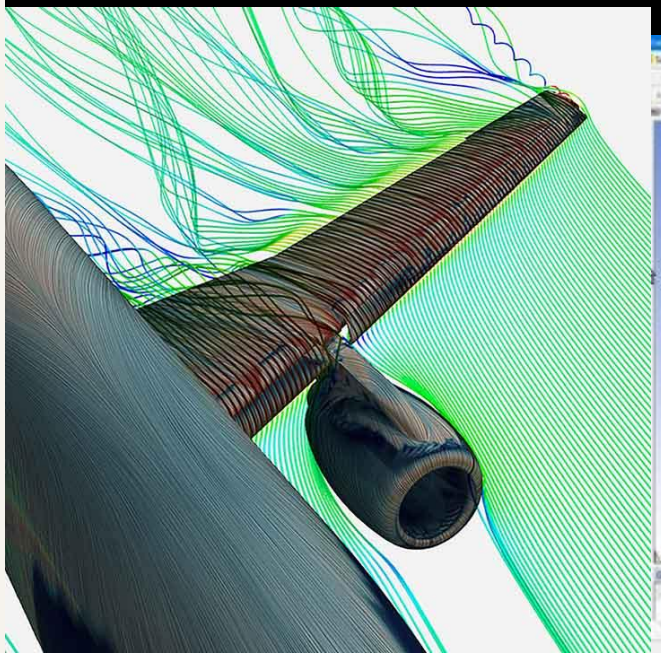
CAE (Computer-Aided Engineering) is the use of computer software to simulate, analyze, and validate the performance of a design or engineering system. It is often used after CAD to test how a product will behave under real-world conditions—such as stress, heat, vibration, fluid flow, or electromagnetic fields—without building a physical prototype.

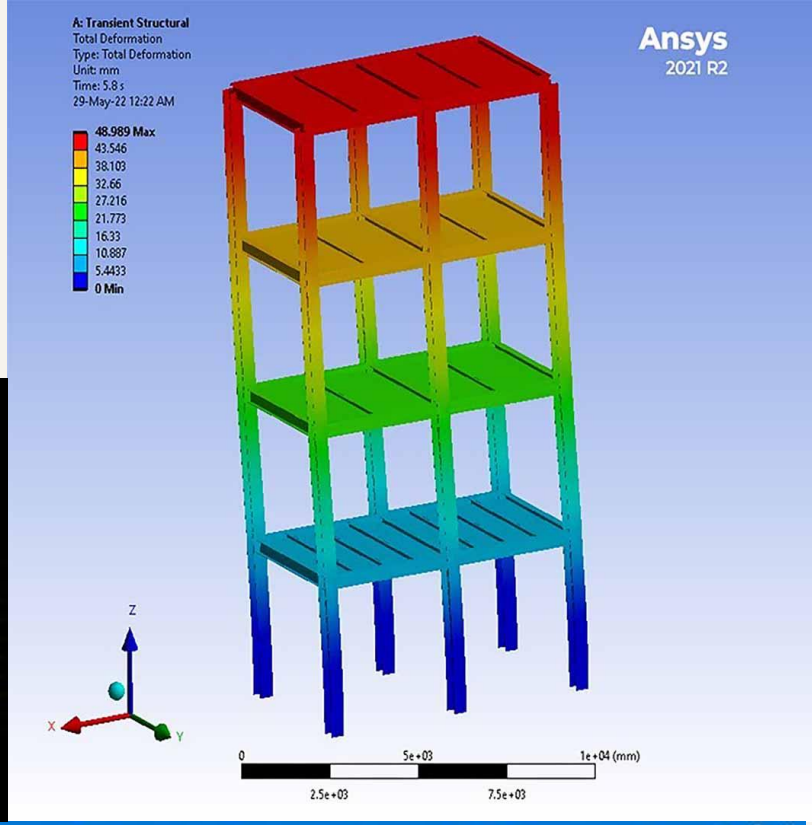
- ANSYS
- COMSOL Multiphysics
- Abaqus
- Altair HyperWorks (Siemens)
- SimScale
- Nastran
- Creo
- SolidWorks Simulation
- Adams
- Simpack
- RecurDyn
- LS-Dyna
- OpenFOAM
- CalculiX
- Radioss
- IcePak

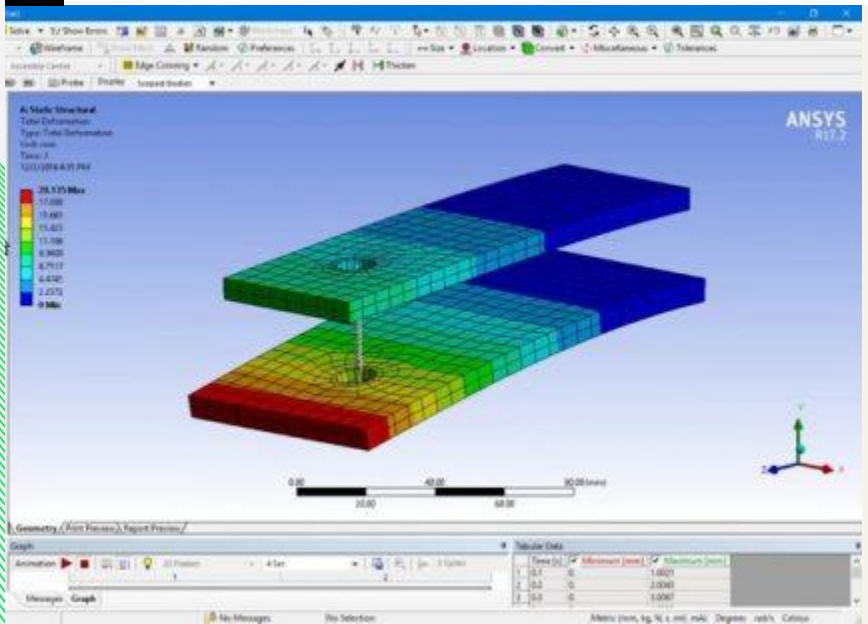
Mechanical Engineering in the Digital Age











Mechanical Engineering in the Digital Age-CNC and

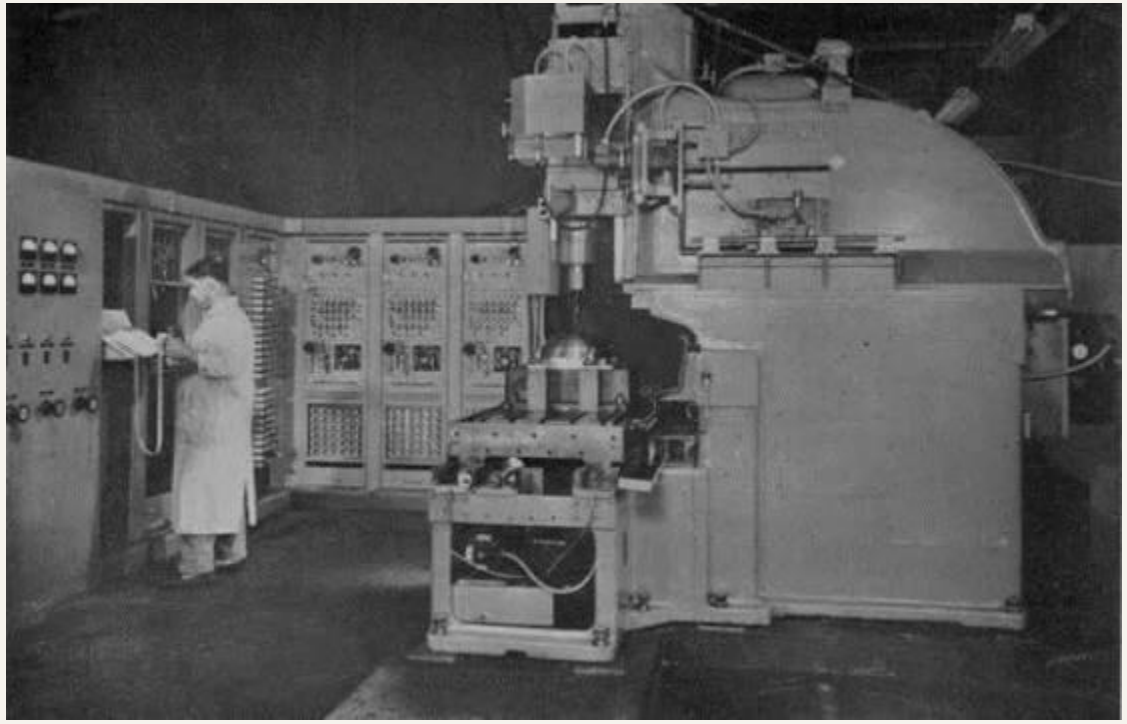
CNC (Computer Numerical Control) is the computerized control of machine tools that follow programmed instructions to manufacture parts with high precision.





Mechanical Engineering in the Digital Age-CNC and

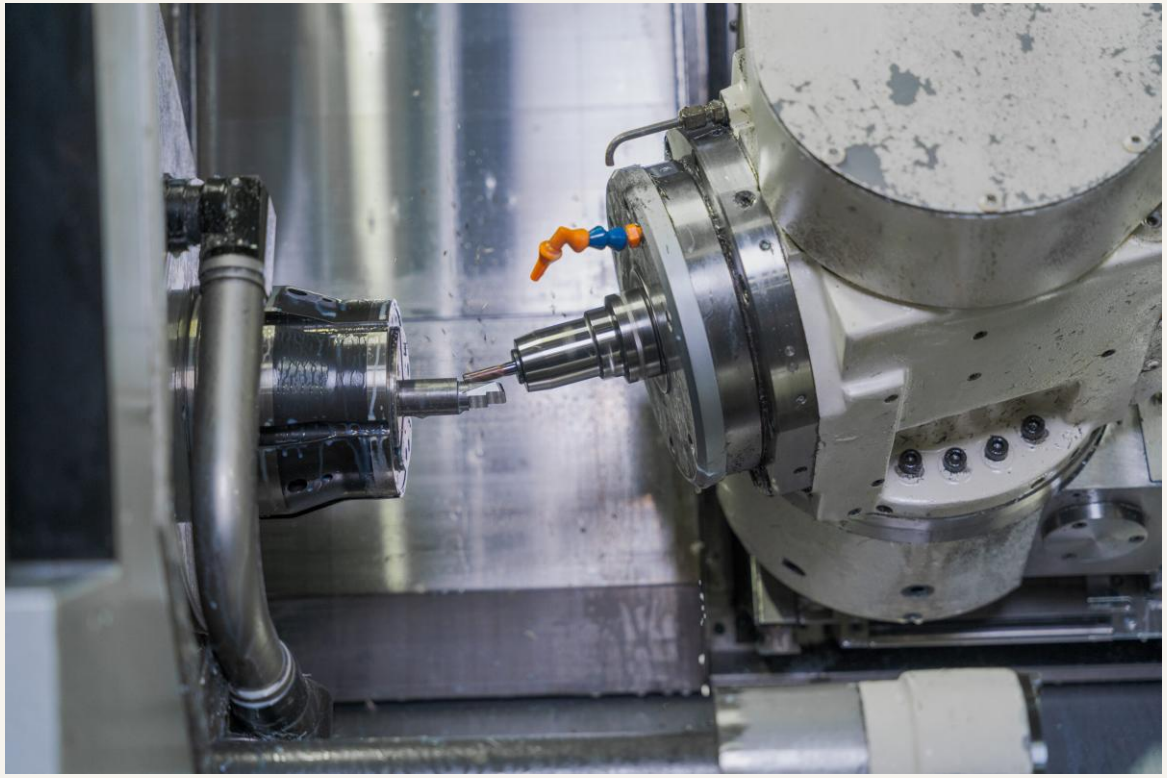
The first CNC was developed in MIT.

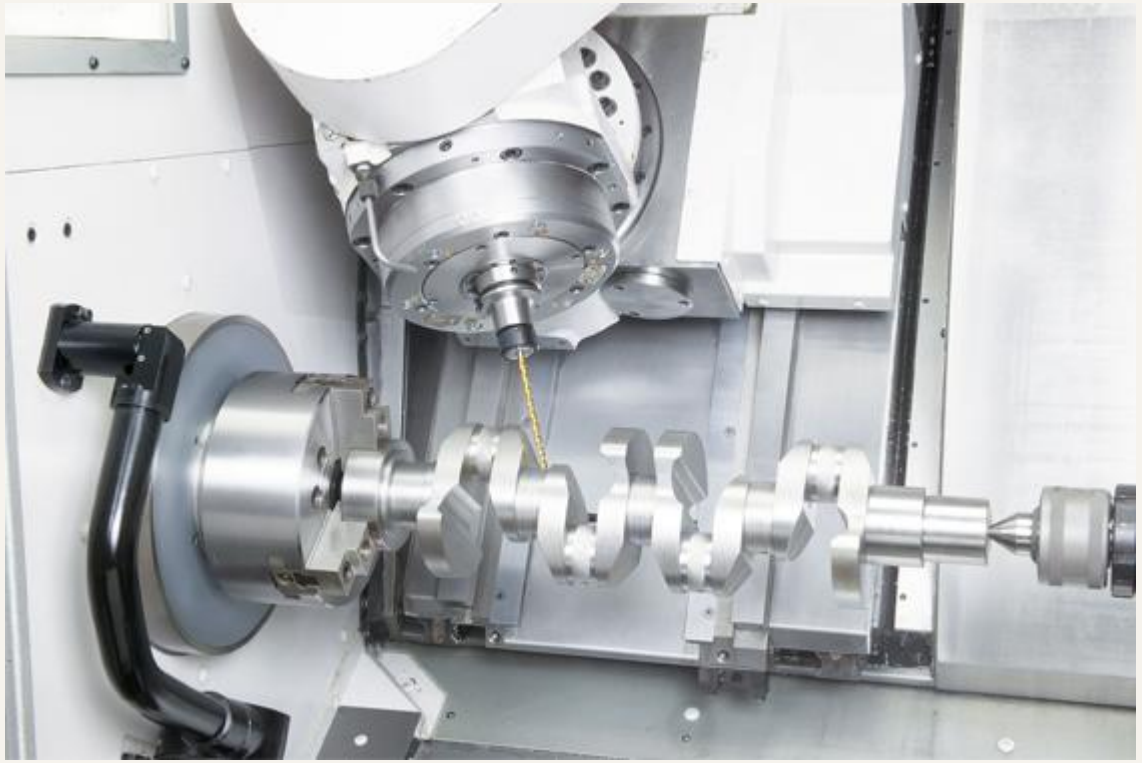




Mechanical Engineering in the Digital Age-CNC and

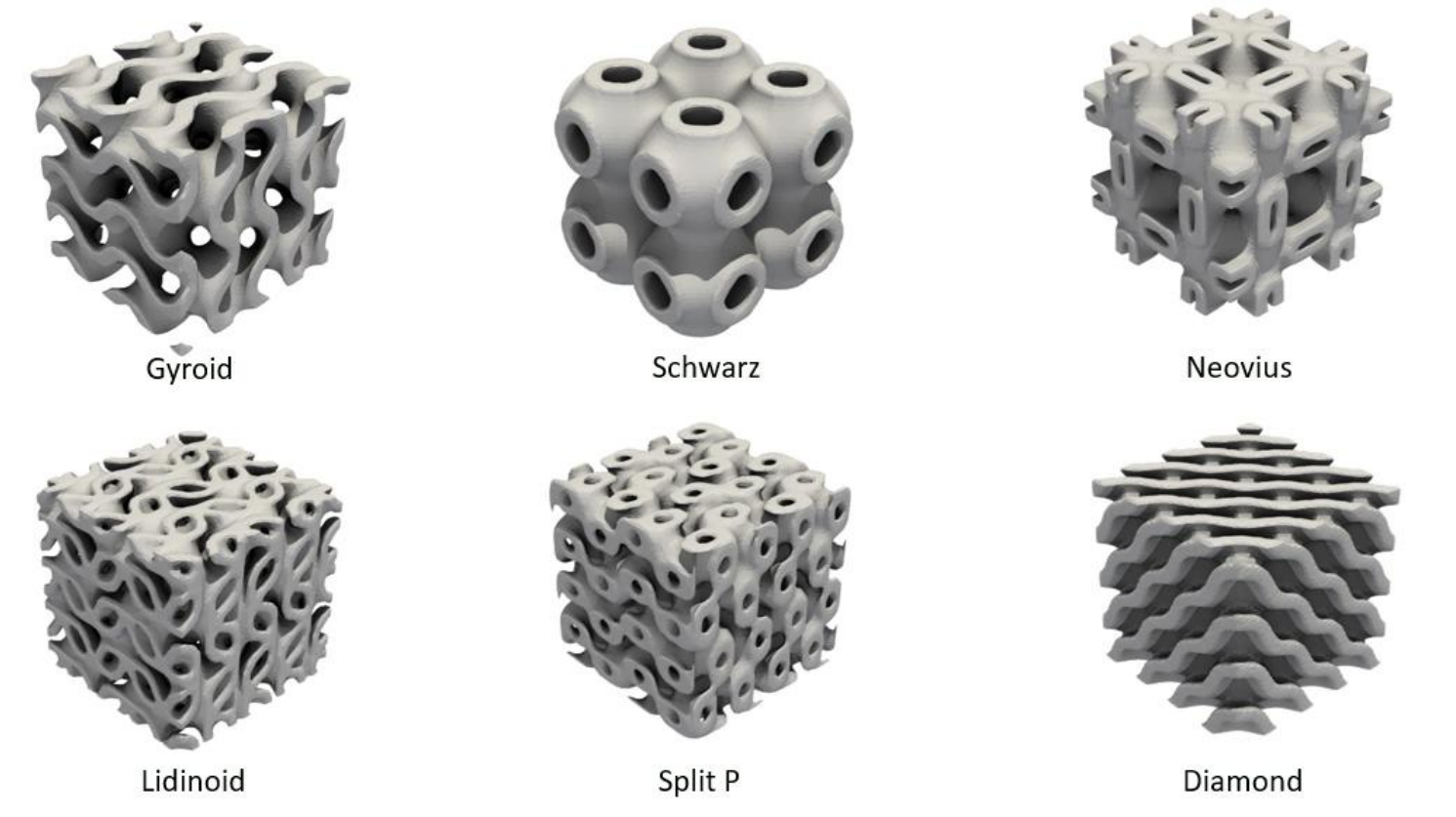
Today’s CNC Machines





Mechanical Engineering in the Digital Age-CNC

Limitations of Conventional Manufacturing



Such geometries are still limitations for conventional manufacturing methods.

3D Manufacturing methods must be used for such geometries.

Mechanical Engineering in the Digital Age

- Why do we still need to learn Math, Physics and Chemistry?
- Why must an engineer still understand Math, Physics and Chemistry?
- Can a software or an AI tool be used instead of pushing engineers and engineering students to learn math, physics, chemistry?

Mechanical Design and Machine Elements

Mechanical design combines analysis with decisions about geometry, materials, assembly, and service.

Common machine elements include shafts, bearings, gears, springs, fasteners, and seals. Each has a specific function, but their interactions determine how the complete machine behaves.

For example, a shaft design must account for transmitted torque, bending loads, bearing locations, fatigue, and the method of manufacture.

Manufacturing Engineering

Manufacturing turns a design into a physical product.

Casting and forming create shapes by solidification or plastic deformation.
Machining removes material to produce features and accurate fits.
Joining connects separate parts.
Additive manufacturing builds a part from deposited or solidified material.

Design and manufacturing must be considered together. A shape that is easy to model may be difficult or expensive to produce.

Measurement, Testing, and Quality

A calculation predicts how a design should behave. A test shows how a particular specimen or prototype actually behaves.

Engineers measure quantities such as force, displacement, temperature, pressure, vibration, and flow rate. They must also consider instrument accuracy and repeatability.

Quality control checks whether manufactured parts meet specified requirements. Testing and inspection provide evidence that a design can perform its intended function.

The Engineering Design Process

A typical design develops through repeated steps:

1- Identify the need.
2-Define measurable requirements.
3-Generate possible concepts.
4- Estimate and compare performance.
5-Build a model or prototype.
6- Test it against the requirements.
7- Revise the design.

Testing may reveal a problem with the concept, the calculations, the manufacturing process, or even the original requirements.

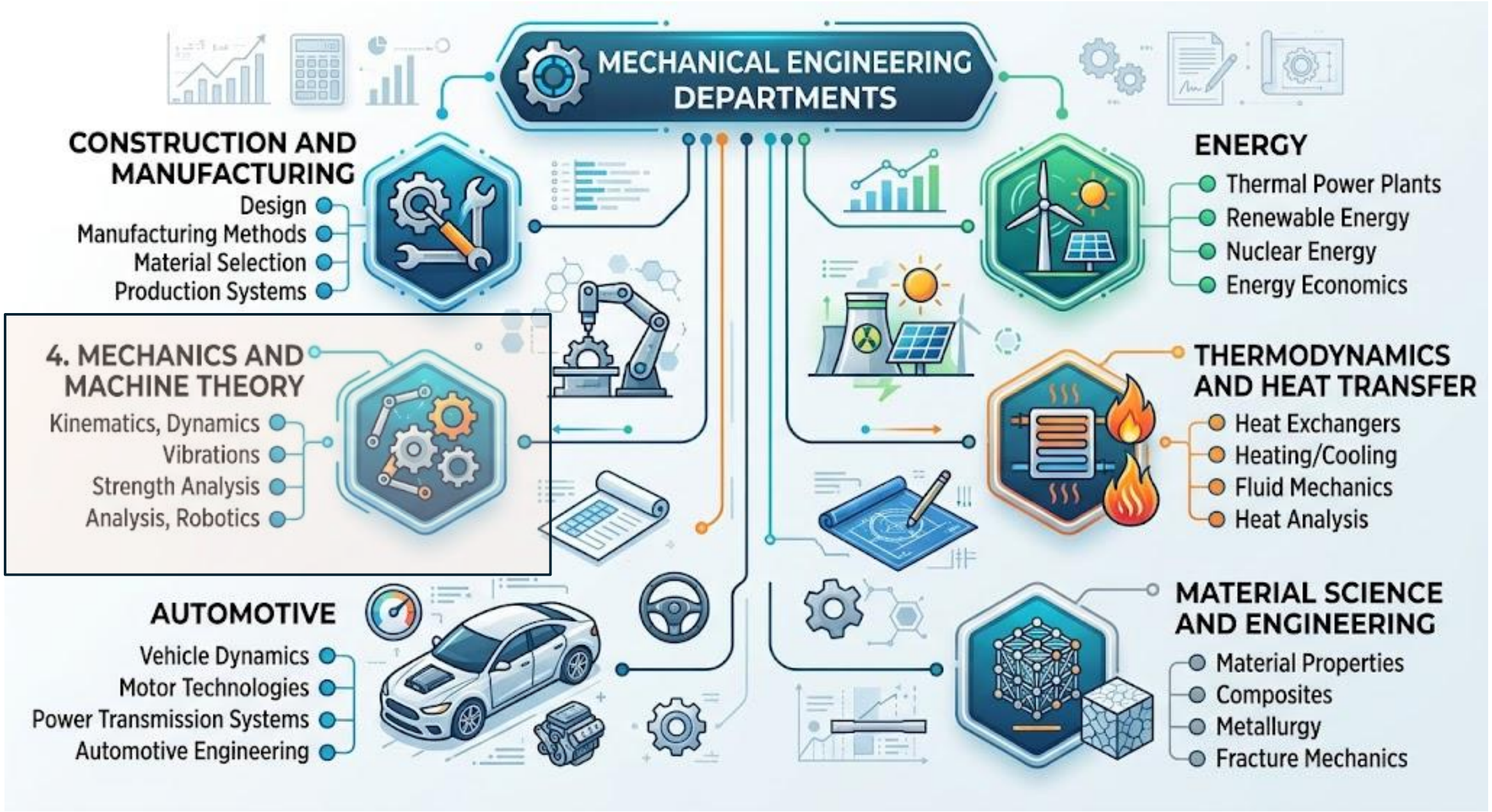
Requirements and Trade-Offs

Engineering problems rarely have a single perfect answer.

A lightweight component may cost more to manufacture. A larger heat exchanger may improve cooling but require more space. A stronger structure may add mass.

Before selecting a design, engineers must decide:

- Which requirements are essential?
- How will performance be measured?
- Which compromises are acceptable?
- What happens if a component fails?


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Engineering Mechanics

Engineering mechanics examines forces and motion.

Statics studies systems in equilibrium.

Dynamics studies motion and the forces that produce it.

Mechanics of materials examines stress, strain, deformation, and failure.

Examples include calculating the load on a bridge support, the acceleration of a robotic arm, or the deformation of a bicycle frame.

Materials and Mechanical Behavior

Material selection determines how a component responds during service.

- Engineers consider:
- Strength and stiffness
- Fatigue and wear
- Operating temperature
- Corrosion resistance
- Mass, availability, and cost

The appropriate material depends on the application. A material suitable for a lightweight housing may be unsuitable for a high-temperature turbine component.

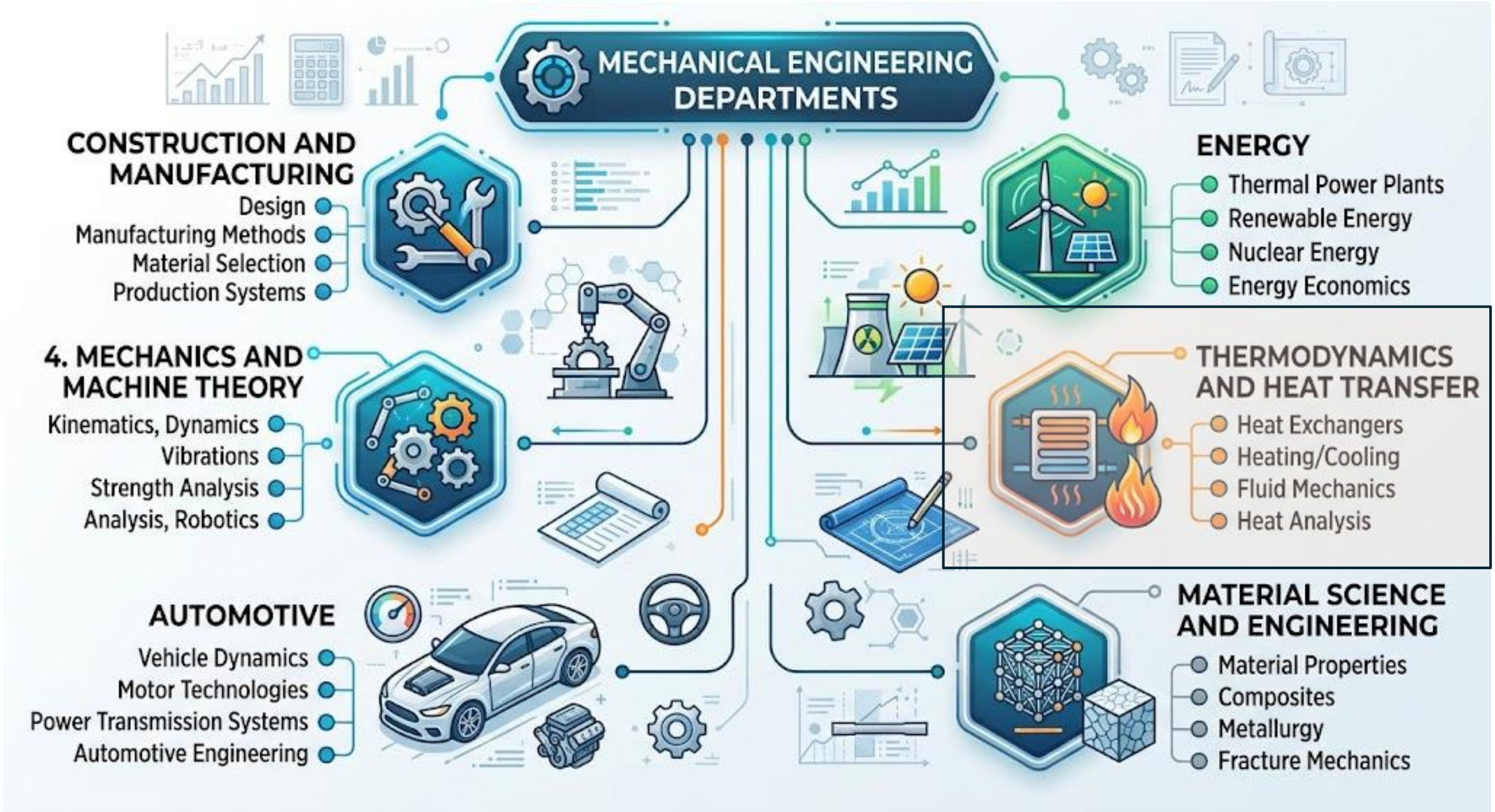
Controls, Automation, and Robotics

Many mechanical systems respond to their environment through feedback.

A basic control system contains:
A **sensor** that measures a condition
A **controller** that compares the measurement with a target
An **actuator** that changes the system

A room thermostat is a familiar example. More complex examples include industrial robots, automated production lines, and vehicle stability systems.

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Thermodynamics and Heat Transfer

Thermodynamics examines energy and its conversion. Heat transfer examines how thermal energy moves between bodies and through materials.

Mechanical engineers apply these subjects to:

- Engines and turbines
- Refrigerators and heat pumps
- Industrial furnaces
- Electronic equipment cooling
- Building heating and ventilation

A thermal system must provide the required performance while using energy responsibly.

Fluid Mechanics

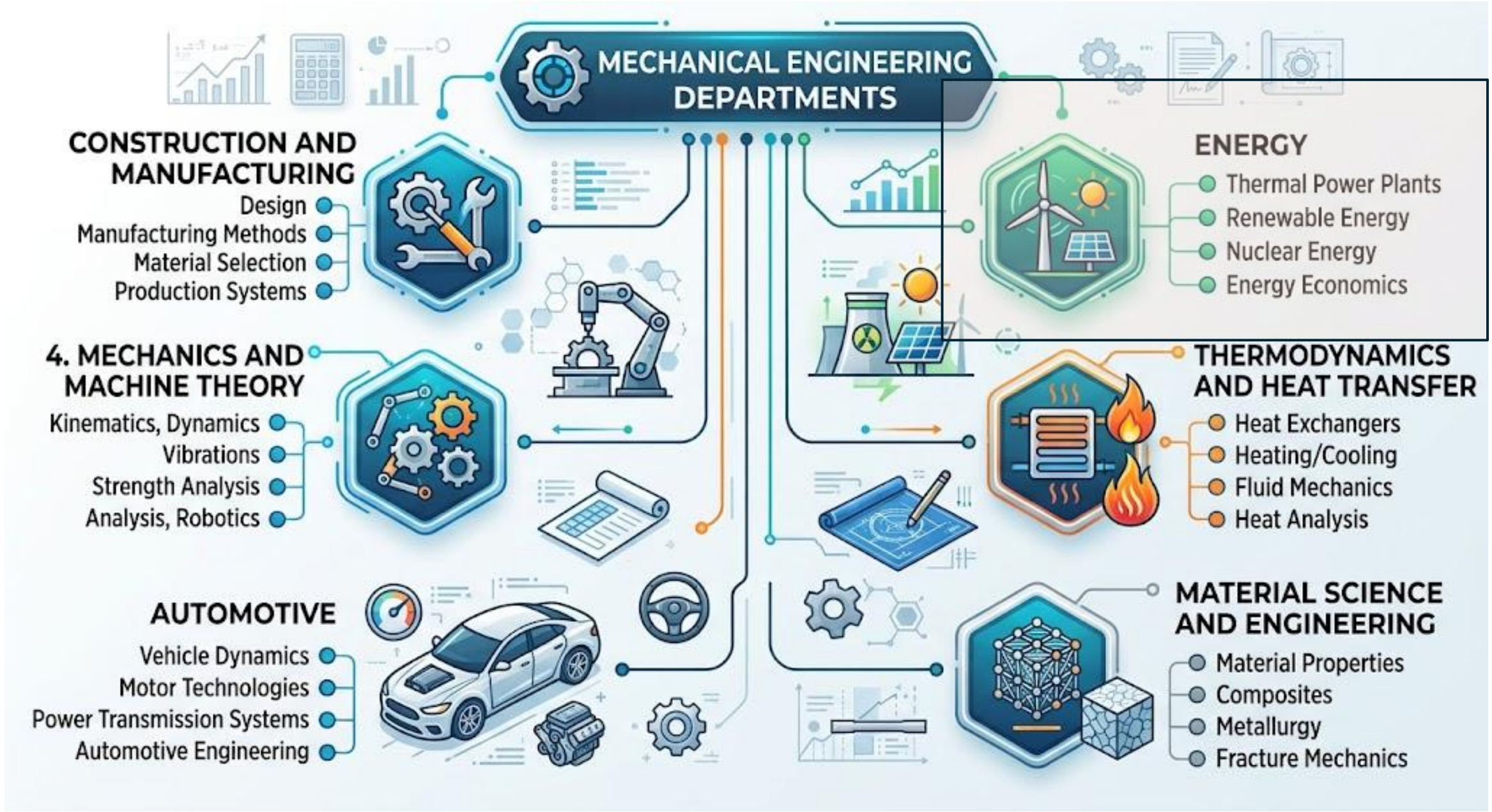
Fluid mechanics studies liquids and gases at rest and in motion.

Mechanical engineers use it to understand:

- Water flow through pipes and pumps
- Airflow through fans and ventilation ducts
- Aerodynamic forces on vehicles and aircraft
- Cooling flow through machinery
- Pressure losses in fluid systems

A change in geometry can alter flow resistance, heat transfer, noise, and energy consumption.

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Energy systems

Power generation, conversion, and use



Energy Systems

Mechanical engineers contribute to the generation, conversion, storage, and use of energy.

Examples include wind turbines, hydroelectric turbines, thermal power equipment, engines, compressors, and heat pumps.

An energy system must be evaluated as a whole. Its useful output, efficiency, operating conditions, maintenance needs, and environmental effects all matter.

MECHANICAL ENGINEERING DEPARTMENTS

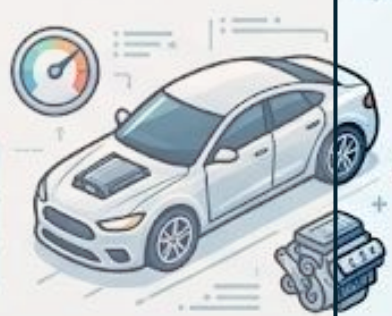
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Transportation Systems

Vehicles combine many branches of mechanical engineering.

- Structures carry passengers and equipment.
- Suspensions and tires influence handling.
- Brakes convert motion into heat.
- Powertrains deliver torque to the wheels.
- Cooling systems protect components.
- Aerodynamic design affects energy use and stability.

A change to one subsystem can influence the performance of others.

Aerospace and Defense Applications

Aircraft and other aerospace systems require careful control of mass, loads, heat, vibration, and reliability.

Mechanical engineers may work on structures, propulsion components, thermal systems, mechanisms, or manufacturing processes. Testing is particularly important because operating conditions can differ greatly from those on the ground.

The engineering principles are familiar; the performance requirements are often demanding.

Biomedical and Biomechanical Applications

Mechanical principles also help engineers study and support the human body.

Applications include artificial joints, prosthetic devices, surgical instruments, rehabilitation equipment, and the analysis of movement.

A biomedical design must consider mechanical performance together with the needs of patients and clinicians. For example, a prosthetic component must carry load while remaining comfortable and suitable for its user.

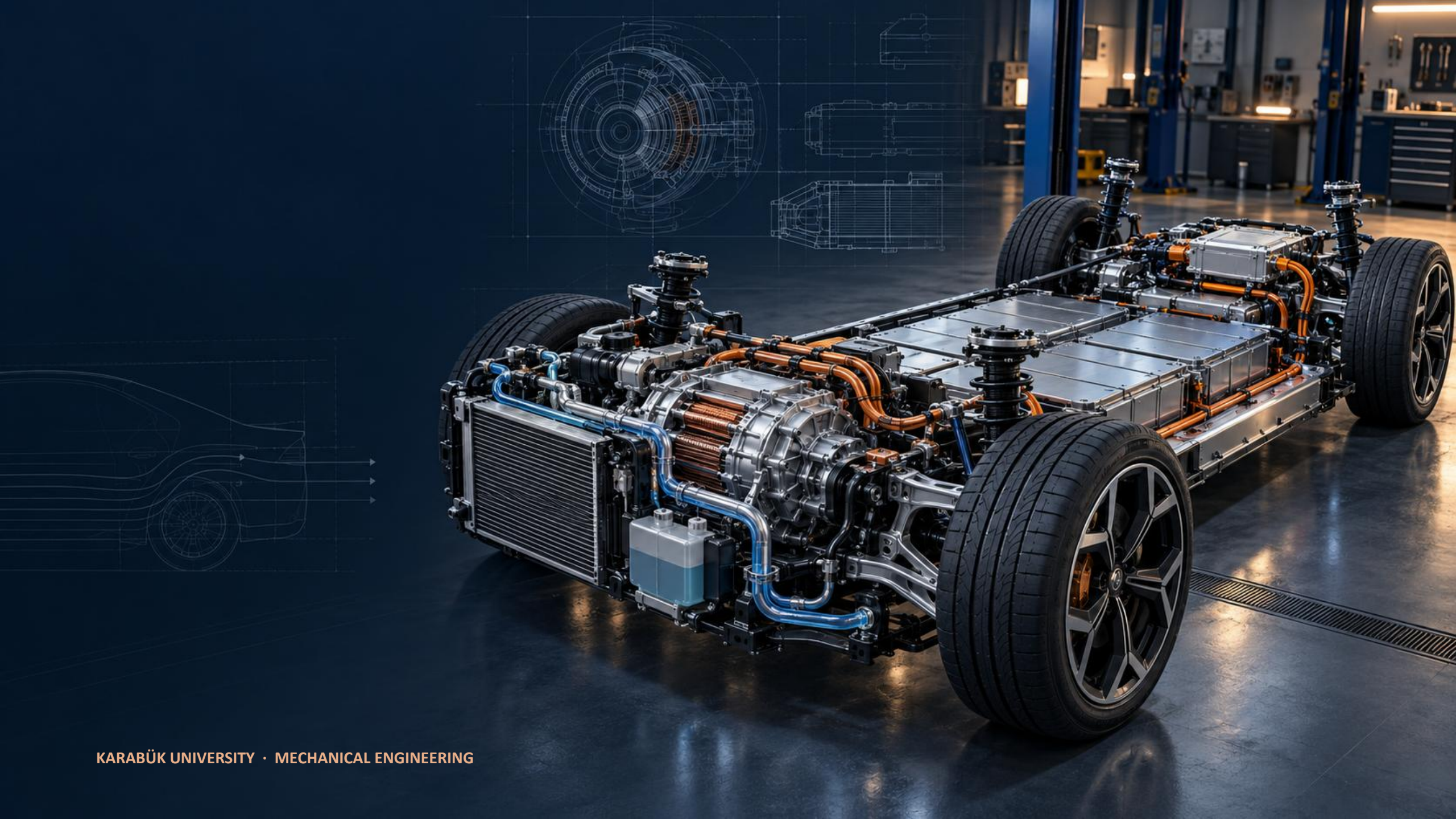
https://biomicrosystems-me.metu.edu.tr/en/research

Acoustics and Vibration

Moving machines generate vibration. Vibration can produce noise, reduce precision, cause discomfort, or contribute to fatigue failure.

Mechanical engineers study where vibration originates, how it travels, and how it can be measured and controlled.

Examples include vehicle noise, rotating machinery, building equipment, and medical devices. The aim may be to reduce unwanted vibration or, in some applications, to produce a controlled vibration.



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